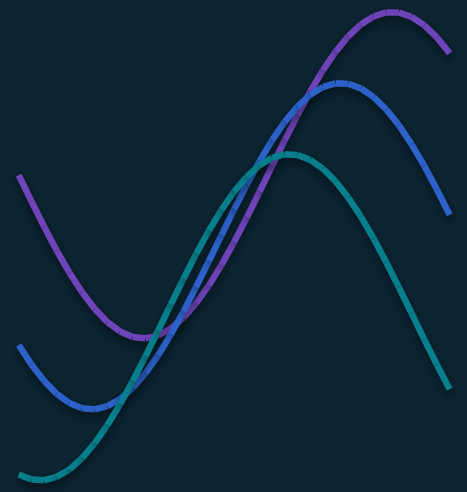
AFS Assessment Assistant



From a customer enquiry to a reviewable first feasibility experiment.

SOURCE-GROUNDED | EXPERT-REVIEWABLE | EXPLICIT UNCERTAINTY

PROJECT TEAM

Eisa Atif • Marie Habermann • Atfan Deshmukh • Akshdeep Singh Naumanurrahman Shaikmujeeburrahman

github.com/ght007/SchubKI • 18 October 2026

The first decision is what to test.

A process goal is not yet a spectroscopy specification.

THE CUSTOMER KNOWS

“We need to monitor moisture or contamination.”

Still unknown: target range, sample optics, references, installation and success criteria.

Clarify the problem

Extract requirements and ask the missing questions that change the decision.

Propose a sensing concept

Recommend methods, spectral range and a high-level probe concept, with sources.

Plan the first experiment

Make uncertainty, practical comparisons and the decision basis explicit.

A fixed pipeline keeps reasoning reviewable.

Language models read and explain; explicit code checks constraints, evidence and confidence.

LLM language | Deterministic code | Checks + one repair

01
Read the enquiry
Facts, unknowns and premises

→

02
Rules + coverage
Method, fiber and scope checks

→

03
Retrieve evidence
BM25 + dense → RRF → MMR

→

04
Test premises
Supported, contradicted or unknown

05
Draft the report
Fast model or Council drafts

→

06
Verify claims
Quotes, numbers and client facts

→

07
Quality gate
16 checks; at most one repair

→

08
Confidence + coach
Signals, caps and follow-up needs

Browser + live SSE • Python / FastAPI / Pydantic • Groq • Supabase + local SQLite fallback

README §§3–8, 21; backend/app/engine/pipeline.py

AFS knowledge becomes checkable evidence.

The source, page, quote and evidence scope travel with the recommendation.

95 curated evidence cards

35 sourced screening rules

16 report quality checks

Evidence with different scopes

S01 + table	Method principles, applications, limitations
S02	Company slides, fibers, bands and AFS paths
S03 / S04	In-vitro proof of concept / laboratory work
S05 / S06	Business and laboratory context

Page → quote → claim → verification

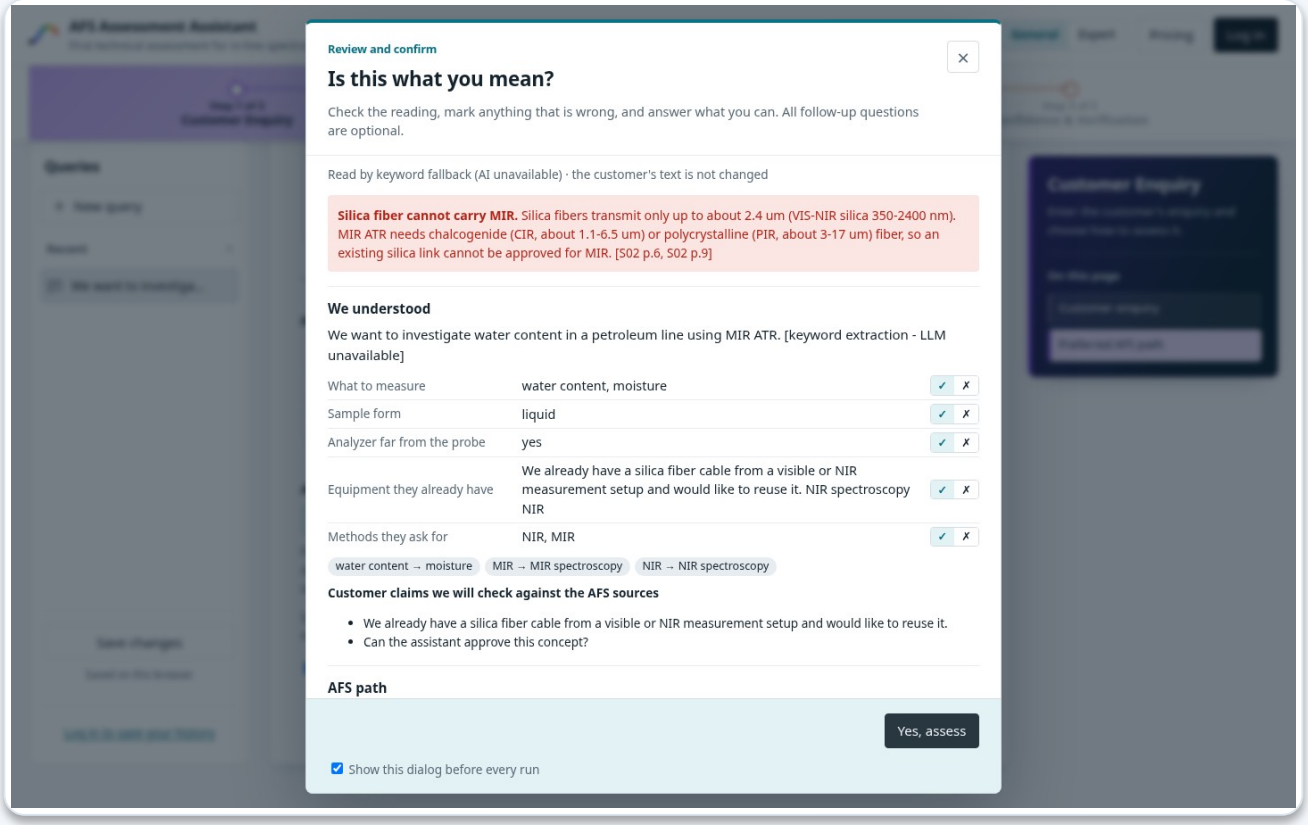
Ingest refuses evidence cards whose quotes cannot be found on their cited pages.

A laboratory or in-vitro result does not establish deployment or clinical readiness.

data/kb/claims.jsonl; rules.yaml; README §§5–7, 21

Review once. Assess with the expert in control.

General: Customer Enquiry → Result Assessment → Confidence and Verification



1 Enter the enquiry

Instant pre-check highlights vague facts and early constraints.

2 Confirm the reading

A dialog contains optional follow-ups. “Yes, assess” submits the answers.

3 Inspect the result

Review the 11-section report, citations, confidence and exported plan.

Current feature_dialog UI; docs/frontend-integration.md; illustrated replay using stored H04

H04: the requested MIR setup cannot be approved.

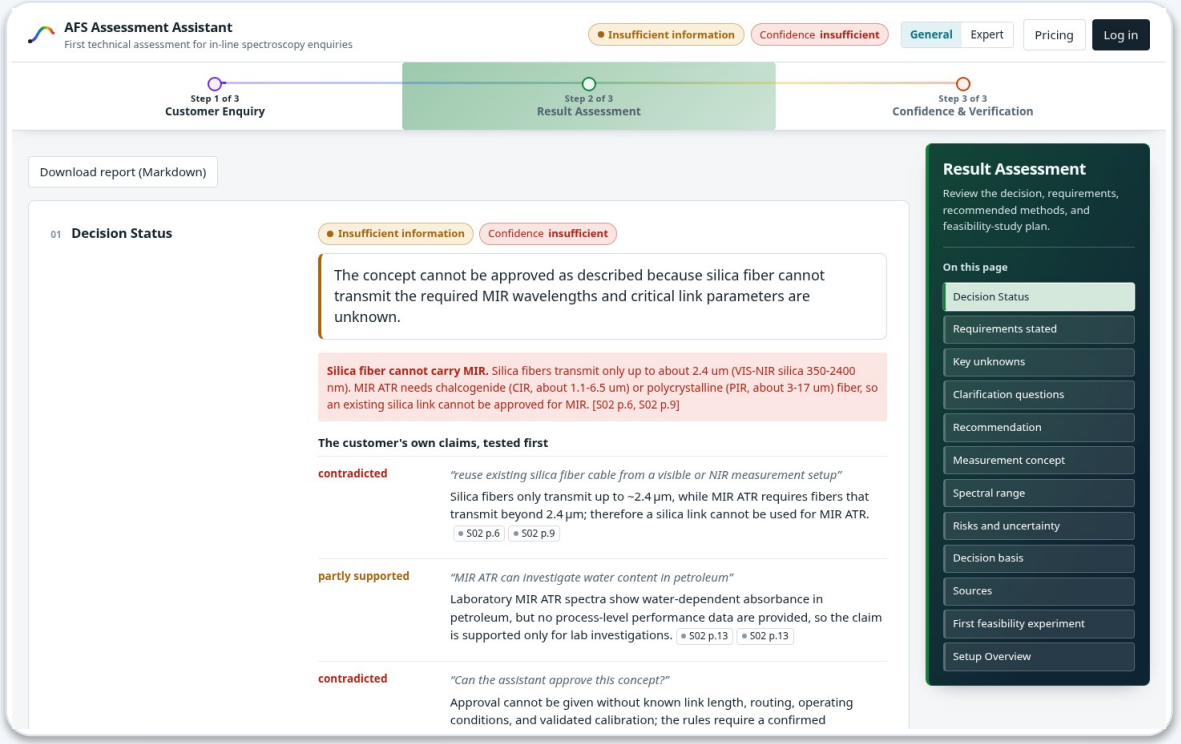
Water in a petroleum line; analyzer in another building; an existing VIS/NIR silica cable.

The assessment

Silica is incompatible with the requested MIR wavelengths.

Unknown: link length, routing and operating conditions.

First test: an IR-compatible link and alternative sensing concepts.



STORED H04 RESULT • INSUFFICIENT INFORMATION

outputs/H04.json; supplied presentation p.5; AFS reference S02 pp.6, 9, 13 via curated KB

A useful recommendation ends with a test.

H04 example: a bench comparison before a remote process installation.

Samples

Water-spiked petroleum spanning relevant process states.

Reference

Independent water measurements, such as Karl Fischer.

Comparison

MIR ATR and alternatives; test the effect of link length.

Decision basis

Client-agreed error, signal and robustness criteria.

AFS lab trial →

Path A

Chemometric model for process automation

Path B

Multispectral system for the R&D laboratory

Path C

Spectral sensor solution for process monitoring

outputs/H04.json, feasibility_plan; challenge brief pp.1–2; AFS pathway via README / S02 p.20

A plausible report can still contain a bad citation.

Verification is visible evidence for review—not a guarantee that every recommendation is correct.

A REAL FLAG IN STORED H04

spectral_range[1]

Unverified:

S12 p.7 does not exist

Stored run	H01	H04
Claim checks[1]	10 / 10	2 / 7
Unsupported numbers	0	1
Quality checks	14 / 16	14 / 16
Confidence	Low	Insufficient

[1] App metric includes verified/fuzzy claims. Two stored examples, not an aggregate benchmark.

One repair round is allowed. Remaining warnings stay visible for expert review.

Stored Fast runs outputs/H01.json and outputs/H04.json (17 Oct 2026); README §§7, 21

Clear boundaries make the assistant useful.

A first technical assessment supports the next conversation and the next experiment.

No measured spectra

Accuracy and detection limits cannot be predicted from the supplied documents.

Next: collect trial spectra, compare models and validate on independent samples.

Curated knowledge

Rules are conservative; unusual wording or missing sources can limit coverage.

Next: expert-approved source additions and feedback-informed cards and rules.

Model and runtime limits

A model can mis-cite; Council agreement is not independent validation and can be slow.

Next: visible verification, Fast-mode demos and broader independent evaluation.

Better first conversations. Checkable next steps.

AFS Assessment Assistant

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Source-grounded recommendations • Explicit uncertainty • Practical feasibility experiments

The deck maps to every evaluation criterion.

Weights reproduced from the official challenge brief; no jury score is claimed.

Criterion	Weight	Evidence in the main presentation
Technical quality	30%	Constraint-aware method and probe concept; H04 example (3, 6).
Use of AFS knowledge	20%	Scoped corpus, citations and source checks (4, 8).
Feasibility-study plan	15%	Samples, references, comparisons and decision basis (7).
Clarification & uncertainty	15%	Review dialog, missing facts and computed confidence (5, 8, 9).
Explainability	10%	Auditable pipeline and visible verification defects (3, 8).
Prototype & presentation	10%	Current UI, timed demo and reproducible repository (5).

Challenge brief p.2; submission form pp.2–4

One assessment, eleven review sections.

The page and Markdown export share the same outline; Setup Overview is separate.

01 Decision Status

02 Requirements stated

03 Key unknowns

04 Clarification questions

05 Recommendation

06 Measurement concept

07 Spectral range

08 Risks and uncertainty

09 Decision basis

10 Sources

11 First feasibility experiment

Separate Setup Overview diagrams illustrate the proposed Path A / B / C systems.

A complete submission needs more than slides.

Checklist from the supplied form. Listed items are requirements, not a claim of submission.

Required item	Handoff / check before submitting
Public repository	Code, setup, models/tools, material-loading instructions and architecture. Verify public access.
Six generated answers	Publish H01–H06, unedited and without case-specific hard-coding.
Accessible slide PDF	Use the companion PDF; publish it with access for the jury.
Video: max. 5 minutes	Demonstrate questions, methods, sources and the experiment; explain limits and architecture.
All five team members	Enter all names and their registered emails in the same order; credits are on slides 1 and 10.

Checkout check: only H01 and H04 output pairs are present. Complete and verify the remaining four before submission.

Submission form pp.1–4; deadline stated there: 18 Oct 2026, 23:59 CEST

Built from the supplied materials and current code.

Current repository: ght007/SchubKI • feature_dialog @18e4f84

1 Official challenge brief

AI-based molecular media analysis by process-spectroscopy in-line.pdf, pp.1–3.

2 Submission requirements

Art Fiber Systems: AI Assistant for In-line Spectroscopy Solution Design.pdf, pp.1–4.

3 Earlier project presentation

AFS_Assistant_Presentation.pdf, 8 pages; used as background, with claims checked.

4 Project documentation

README.md and docs/frontend-integration.md: approach, UI, evaluation, limits and storage.

5 Implementation and evidence

Current pipeline, presenter and render code; data/kb/claims.jsonl and rules.yaml.

6 Stored examples

outputs/H01.json and H04.json, created 17 Oct 2026; screenshots replay those archived data.

Reproduce through the repository README with the required raw AFS documents. Slides alone do not establish runtime readiness.